

Research Idea: BHR_71, HH_212, IRAS_0416+2706,
L1551_IRS_5, CG_30, HH_46, GAL_331.5-00.1,
IRAS_04166+2706

ALMA Thesis Planner (automated pipeline)

The BHR 71 Band 7 spectral scan: a position-resolved chemical inventory of one ACA pointing, built against the primary-beam noise map

Motivation. ALMA project 2017.1.01053.S (SMORES, PI McGuire) has **no publications linked in the archive**, and BHR 71 is its most favourable target: it was observed in **all six frequency tunings** (member OUS X1001, Xfa2, Xfb5, Xfc8, Xfdb, Xfee), giving continuous Band 7 coverage over 292.7–316.0 and 337.7–361.0 GHz on a single field — an unbiased spectral scan, not a handful of lines. The proposal’s central claim is *spatio-chemical differentiation*: that SiO (grain-core sputtering in fast J-shocks), CH₃OH (ice-mantle release in slower C-shocks at cavity walls), and quiescent envelope gas occupy distinct physical zones rather than one well-mixed reservoir. That claim is about **position**, so the thesis measures it at fixed positions — but only at as many positions as one 7-m pointing can honestly deliver.

The project.

1. **Establish the usable field before choosing anything.** Mosaic = "no": this is a *single* ACA pointing, FWHM $\approx 29''$ at 345 GHz and $\approx 34''$ at 300 GHz, while BHR 71’s outflow extends over arcminutes. **Step one of the thesis is the primary-beam-corrected noise map for each tuning.** Every aperture is then selected against measured local RMS, not against a nominal tabulated sensitivity: at $r \approx 12\text{--}15''$ the PB-corrected noise is roughly double the tabulated 8–14 mJy/beam per 10 km/s, and an upper limit is only worth reporting if its noise is the real one. The deliverable of this step is a stated *usable radius* inside which all science apertures must fall.

2. **Continuum, budgeted as real work.** Only cube products are archived, so continuum images come from per-tuning, per-pixel fitting of line-free channels (or imaging the MS in CASA). Toward a chemically rich source in a spectral scan there may be very few genuinely line-free channels per tuning; the thesis reports the fraction of channels used, and states that the continuum peak — which anchors the aperture scheme — carries residual line contamination at a quantified level.

3. **Homogenise to 4.2''.** BHR 71’s own beams span 3.53''–4.15'', so the common circular beam is set by *this field*, at $\sim 4.2''$ — not by other targets in the programme. Smooth with ‘imsmooth’, primary-beam correct, convert to brightness temperature. Every number comes from matched-beam cubes.

4. **Three zones, not eight apertures.** At 4.2'' inside a $\sim 29''$ FWHM pointing there are only $\sim 5\text{--}6$ independent resolution elements across the whole usable field. The analysis therefore reports **three primary apertures**, chosen by stated data-internal criteria and held fixed for every line: (a) the IRS 1 continuum peak; (b) the brightest SiO 8–7 (347.331 GHz) position offset from it

along the outflow axis; (c) the most offset position where CO 3–2 (345.796 GHz) shows a narrow, near-systemic profile and SiO is absent, subject to the noise map from step 1. **Positions are spatially correlated and this is stated up front: no per-position significance statistics are run, and only contrasts between genuinely separated zones are reported.** Aperture (c) is labelled honestly — if the noise map shows it must sit within the inner envelope, it is called *inner envelope*, not *ambient*, and the framing stays three-zone rather than being inflated to four.

5. Line inventory as a standalone deliverable. A full identification table across all six tunings, **every rest frequency and E_{up} verified in Splatalogue** — the line list in the data description is explicitly LLM-generated and is a starting point only. An unpublished ACA spectral scan of a well-known outflow field is a result on its own, independent of whether any downstream fit succeeds.

6. Excitation, with its dominant systematic named. CH₃OH rotation diagrams per aperture, anchored on the 338 GHz K-ladder, which **lies within a single tuning** and is therefore largely immune to the 10–20% inter-tuning flux-scale error. The diagram will work at the continuum peak; it will likely fail at the SiO position and the offset reference. **The T_{ex} used for N(SiO) at the shock position is therefore an assumed band, and this — not calibration — is the leading systematic on the headline N(SiO)/N(CH₃OH) contrast.** The band is stated numerically and propagated; the contrast is reported as a range across it. Toward a hot-corino-like peak the bright 338 GHz lines may be optically thick and beam-diluted, biasing the diagram low, so **¹³CH₃OH transitions falling in the coverage are checked explicitly, at minimum as an optical-depth limit.** LTE columns follow for SiO (303.927, 347.331 GHz), SO (304.078, 344.311 GHz), H₂CO (351.769, 355.996 GHz), CS (293.912, 342.883 GHz). CO 3–2 is used for morphology, velocity extent, and aperture definition only. SiO 7–6/8–7 is an optical-depth and cross-tuning consistency check, never a thermometer.

7. The measurement. Zone-to-zone abundance contrasts — N(SiO)/N(CH₃OH), with SO, H₂CO, CS referenced to both. Kinematic decomposition against the systemic velocity from H¹³CO⁺ 4–3 (346.998 GHz) is a **second axis applied within each aperture**, testing whether the high-velocity wing at the protostellar peak resembles the shock position or the compact source sharing its beam. This velocity axis is the part that survives regardless of how tight the usable field turns out to be.

Stretch goal, explicitly demoted. IRS 2 sits close enough to IRS 1 to be near the half-power radius, so an internal two-outflow comparison at identical distance, calibration, and beam is the *likely-to-evaporate* element, not the headline. It is attempted only if the step-1 noise map shows photometry there is trustworthy, and the thesis abstract is written to the single-source inventory either way.

Scope. One field, six tunings, three apertures × ~5 species, plus a full line-ID table. A semester of measurement, a semester of writing. HH 212 or CG 30 (four tunings each) is a bolt-on second chapter only if the first closes early.

Honest systematics. (a) T_{ex} assumption at the shock position dominates, as above. (b) Band 7 absolute flux calibration is 10–20% and does **not** cancel across tunings; tunings abut with only ~100–200 MHz of edge overlap, so an empirical cross-tuning check is opportunistic at best — hence the excitation result is confined to one tuning by design. (c) ACA-only *uv* coverage filters extended emission and does so differently across 292–361 GHz; matched-beam smoothing equalises resolution but does not repair this, and it biases the most-offset aperture hardest. (d) At 4.2'' this is aperture spectroscopy of zones; no claim is made about sub-beam structure. (e) Distances are literature values not given in the data description, stated as assumed; no conclusion depends on them, since

all comparisons are internal to one field. (f) CH_3OH may be undetected at the offset reference or in the faintest shocked gas — firm, noise-map-justified upper limits on $N(\text{CH}_3\text{OH})$ where SiO is bright are themselves a direct answer to the differentiation question.